BASIC CIVIL & MECHANICAL ENGINEERING

Subject Code: 23A01201T | UNIT – II : SURVEYING

I. Introduction & Objectives of Surveying

> **Surveying** is the art and science of determining the **relative positions of points on, above or beneath the earth's surface** by measuring horizontal distances, vertical distances (heights) and angles, and plotting them to scale as a **plan or map**.

❖Objectives

- To prepare **plans/ maps** of an area to a suitable scale showing natural & man-made features.
- To determine **areas and volumes** (earthwork, reservoir capacity).
- To **set out** (locate) structures on the ground – buildings, roads, canals, bridges, tunnels.
- To get **levels/ elevations** & contours for design of roads, railways, drainage, dams.
- Fixing **boundaries**, land records, town planning, mining, defence & geological studies.

❖Primary divisions

- **Plane surveying:** earth surface treated as **plane**; area < 260 km^2 – ordinary civil work.
- **Geodetic surveying:** curvature of earth is considered; large areas; high accuracy (national control).

❖Classification (by instruments/ methods)

- **Chain (tape) surveying, Compass surveying, Plane table surveying, Levelling, Theodolite surveying, Tacheometry, Total station / GPS / drone / photogrammetry / remote sensing.**
- By purpose: topographical, cadastral (property), city, engineering, hydrographic, mine, route, astronomical.

❖Fundamental principles

- Working from whole to part (fix big control framework first, then fill in details – prevents error accumulation).
- Locating a new station by at least two measurements from fixed points (linear/ angular).
- Provide checks; take independent measurements; keep field notes neat.

2. Horizontal Measurements – Linear Measurements

- Horizontal distance between two points is measured by direct methods (chain/ tape) or indirect (tacheometry, EDM, total station, GPS).

❖Instruments

- Metric chain: 20 m (100 links, each link 0.2 m) or 30 m; tags every 2 m / 5 m; brass handles. Gunter's chain 66 ft (100 links). Steel band; Invar tape (precise); cloth/ metallic tape (20, 30 m).
- Arrows (marking pins, 10 per set), ranging rods (2 m, painted red-white), offset rod, plumb bob, cross-staff / optical square (to set right angles), pegs, EDM/ laser distance meter.

❖Ranging

> Ranging = fixing intermediate points on a chain line so that the chain is laid in a straight line.

- Direct ranging: by eye with ranging rods when end stations visible. Indirect ranging: by reciprocal ranging when intermediate ground is high.

❖Chaining on level & sloping ground

- Hold the chain horizontal and taut; leader & follower; arrows exchange every chain length; offsets measured perpendicular to the chain line (perpendicular or oblique).
- Slope correction: horizontal length = $L \cos\theta$ (or $C_s = h^2/2L$).

❖Chain survey (simple)

- Area divided into well-conditioned triangles (angles 30°–120°, ideally equilateral), base line, tie lines, check lines; field book records (chainage, offsets).
- Suitable for small, open, flat areas.

❖Errors in chaining & corrections

- Cumulative vs compensating; Instrumental (chain too long/ short), natural (temperature, sag), personal, mistakes (miscounting).
- Correction for incorrect length: True length = $L \times (L'/L)$ where L' = actual length of chain, L = nominal.
- Tape correction: temperature $C_t = \alpha (T_m - T_0) L$; sag $C_s = w^2L^3/(24P^2)$; pull $C_p = (P - P_0)L/(AE)$.

❖Example

> A 20 m chain was found to be 20.05 m long. A line measured 400 m with this chain. True length = 400 × 20.05/20 = **401.0 m**. If the chain is 0.05 m too short: true = 400×19.95/20 = **399.0 m**.

> A line of 150 m on a slope of 5°: horizontal distance = 150 cos 5° = **149.43 m**.

3. Angular Measurements

- Horizontal angles are measured with **compass** (magnetic bearings), **theodolite** (accurate angles), **sextant**, **total station**, and **plane table** (graphically).

❖Theodolite

- Most accurate instrument for **horizontal & vertical angles**; also used for prolonging lines, levelling, tacheometry.
- Parts: telescope, **vertical circle** (graduated), **horizontal circle**, plates, **levelling head** (foot screws, tribrach), plate levels, **vernier/ micrometer**, tripod & plumb bob.
- Transit (vertical-axis, telescope can be rotated 360° in vertical plane); Vernier vs electronic/ digital (least count 20", 10", 1").
- Temporary adjustments: centring, levelling, elimination of parallax.
- Measuring horizontal angle: set up at station O, level; sight left object A (zero), note reading; clockwise to B, read; angle AOB = difference; repeat with face right & take mean to reduce errors (face left & face right observations).

❖Total station

- Electronic theodolite + EDM + microprocessor → gives angles, distance, coordinates (X, Y, Z) simultaneously, stores data; used for large projects, mapping, setting out.
- GPS/ GNSS: satellite based positioning; Drone/ LiDAR surveys for contours/ DEM.

4. Introduction to Bearings

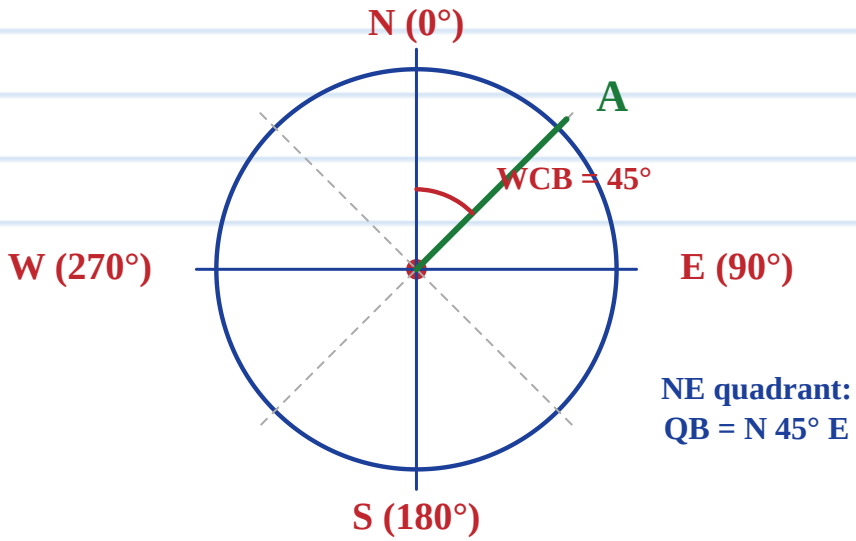


Fig 2.1 Whole circle bearing (WCB, clockwise from North) and quadrantal bearing

> Bearing of a line = horizontal angle between a reference meridian (North) and the line, measured clockwise.

- Meridians: True (geographic north pole), Magnetic (direction of freely suspended needle), Arbitrary.
- Whole Circle Bearing (WCB): measured clockwise 0°–360° from North.
- Reduced / Quadrantal Bearing (RB/ QB): angle from N or S towards E or W, 0°–90° – e.g. N 45° E, S 30° W.

❖Conversion WCB ↔ QB

WCB range	Quadrant	QB
0° – 90°	NE	QB = WCB (N θ E)
90° – 180°	SE	QB = 180° – WCB (S θ E)
180° – 270°	SW	QB = WCB – 180° (S θ W)
270° – 360°	NW	QB = 360° – WCB (N θ W)

❖Fore bearing & Back bearing

> Back bearing (BB) = Fore bearing (FB) ± 180° (+180° if FB < 180°, −180° if FB > 180°)

- FB: bearing in the direction of survey progress; BB: reverse direction.

❖Included angle from bearings

> Included angle at a station = (Back bearing of previous line) ~ (Fore bearing of next line) – i.e. difference between the FB of next and BB of previous line

- Interior angle sum of a closed traverse = (2n − 4) × 90°.

❖Prismatic compass

- Used to measure bearing: **magnetic needle** (balanced on pivot), graduated **aluminium ring/ card** (0° at south end, reading through a prism – WCB), sighting vane (object slit & hair), **prism, lid with mirror**, glass cover, brake pin. Least count 30′.
- **Surveyor's compass:** card fixed to needle; quadrantal bearing; E and W interchanged.

❖Magnetic declination & dip

- **Declination** = angle between true & magnetic meridians (varies place to place & time); **True bearing = Magnetic bearing ± declination (East +).**
- **Dip** = inclination of needle; counter-balanced by a rider (brass coil).

❖Local attraction

- Disturbance of the needle by magnetic substances (iron, steel, power lines). **Detected when FB − BB ≠ 180°. Corrected by the first-line method** or by correction in included angles.

❖Solved problems on bearings

Q1. FB of AB = 62° 30′ → BB = 242° 30′. FB of CD = 215° → BB = 215° − 180° = 35°.

Q2. Convert WCB 285° to QB: in NW quadrant → QB = 360° − 285° = N 75° W. WCB 135° → S 45° E. WCB 215° → S 35° W.

Q3. QB = S 40° E → WCB = 180° − 40° = 140°. QB N 25° W → WCB = 335°.

Q4. FB of AB = 80°, BB of BC (i.e. bearing of CB) = 300°, find ∠ABC. FB of BC = 300° − 180° = 120°; BB of AB = 80° + 180° = 260°. Angle at B = (FB of BC) − (BB of AB) = 120° − 260° = −140° → +360° = 220° (exterior), interior = 140°.

Q5. Magnetic bearing of a line = 125° 30′, declination = 4° 20′ W. True bearing = 125°30′ − 4°20′ = 121° 10′.

Q6. Local attraction: FB of AB = 45° 10′, BB = 226° 30′ → difference 181° 20′ ≠ 180°. Mean = (45°10′ + (226°30′ − 180°))/2 → correct FB = 45° 50′ if station A is free of attraction.

5. Levelling

Levelling is the branch of surveying that determines **elevations (heights)** or **differences in elevation** of points with respect to a **datum**, using a level and staff.

❖Basic terms

- **Datum:** reference surface (**Mean Sea Level – MSL**, Karachi/ Chennai MSL in India).
- **Reduced Level (RL):** elevation of a point above datum. **Benchmark (BM):** fixed point of known RL (GTS BM, permanent, temporary, arbitrary).
- **Line of collimation:** line of sight through the crosshair & optical centre. **Height of instrument (H.I.)** = RL of line of collimation.
- **Back sight (BS):** first reading after setting up level, on a point of known RL. **Fore sight (FS):** last reading before shifting. **Intermediate sight (IS):** any other reading between BS & FS. **Change point (CP / turning point):** point where both FS & BS are taken as the instrument is shifted.
- **Station:** point where staff is held. **Rise & Fall.**

6. Instruments for Levelling

- **Dumpy level:** telescope rigidly fixed to the stand; **bubble tube**, foot screws, **levelling head**, tripod; simplest, most common; adjustments: temporary (setting up, levelling, focusing) and permanent.
- **Tilting level:** telescope can be tilted slightly about a horizontal axis; for quick, precise work.
- **Automatic (self-aligning) level:** compensator keeps line of sight horizontal – fast, popular today.
- **Digital level / laser level / Total station** – bar-coded staff; automatic reading & booking.
- **Y-level, cooke's reversible level** (older types).
- **Levelling staff:** wooden/ aluminium rod, **telescopic or folding**, 3–5 m long, graduated in metres, decimetres, centimetres (E-shaped, 5 mm least count; black & red numerals). Held **vertical** (check by plumb bob/ staff bubble). Foot plate for change points.

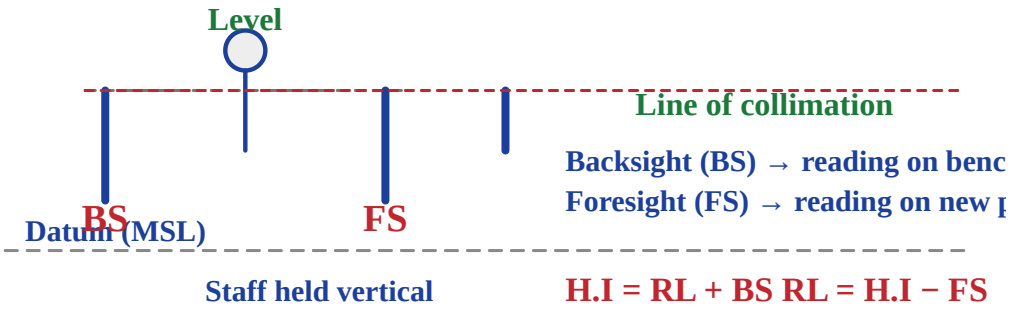


Fig 2.2 Levelling – height of instrument method

❖Temporary adjustments of a dumpy level

- **Setting up** the level (tripod head approx level, firmly fixed). **Levelling up** with 3 foot screws using the plate bubble (two screws parallel to a line, third perpendicular; turn 90°). **Elimination of parallax** (eye-piece focus on cross-hairs, objective focus on staff).

❖Curvature & refraction

- Combined correction = $-0.0673\ d^2$ (d in km, result in m); applied for long sights (> 100–150 m; ignored when BS = FS distances equal – so keep BS and FS distances equal).

7. Booking & Reduction of Levels

❖(a) Height of Instrument (Collimation) method

H.I. = RL of BM + BS ; RL of point = H.I. − FS (or IS) ; New H.I. at change point = RL of CP + new BS

❖(b) Rise & Fall method

- Rise/ Fall = (previous reading) − (current reading); positive = Rise (+), negative = Fall (−). RL = previous RL + Rise or − Fall.
- Arithmetic check: $\Sigma BS - \Sigma FS = \Sigma$ Rise $- \Sigma$ Fall = Last RL − First RL.

❖Which is better?

- HI method – quicker, convenient when many intermediate sights (profile levelling). Rise/Fall – checks every point; used when IS are few.

❖Types of levelling

- Simple (one setup), Differential (many setups between far points), Profile/ longitudinal sectioning (along a road/ canal line at 20–30 m intervals), Cross-sectioning (perpendicular to profile), Reciprocal levelling (across river/ valley – eliminates curvature & refraction), Fly levelling (checking), Check levelling, Precise levelling, Barometric/ trigonometric (indirect).

❖Example 1 – H.I. method

Station	BS	IS	FS	H.I.	RL
BM	1.525			101.525	100.000
A		1.940			99.585
B			2.105		99.420
B (CP)	1.355			100.775	99.420
C			0.885		99.890

RL of A = 101.525 − 1.940 = 99.585; RL B = 101.525 − 2.105 = 99.420; new H.I. = 99.420 + 1.355 = 100.775; RL C = 100.775 − 0.885 = 99.890. Check: $\Sigma BS - \Sigma FS$ = (1.525+1.355) − (2.105+0.885) = −0.110 = 99.890 − 100.000 ✔.

❖Example 2 – Rise & Fall

Readings: 2.285 (BS on BM, RL 200.000), 1.650, 1.895, 2.350 (CP), 1.545 (BS), 0.975 (FS).

Rise 1→2: 2.285−1.650 = +0.635 → RL 200.635; 2→3: 1.650−1.895 = −0.245 → 200.390; 3→CP: 1.895−2.350 = −0.455 → 199.935; after shift: 1.545 − 0.975 = +0.570 → 200.505.

❖Example 3 – Difference in elevation

> BS on A = 1.275, FS on B = 2.610. Difference = 1.275 − 2.610 = −1.335 → B is 1.335 m lower than A.

8. Contouring

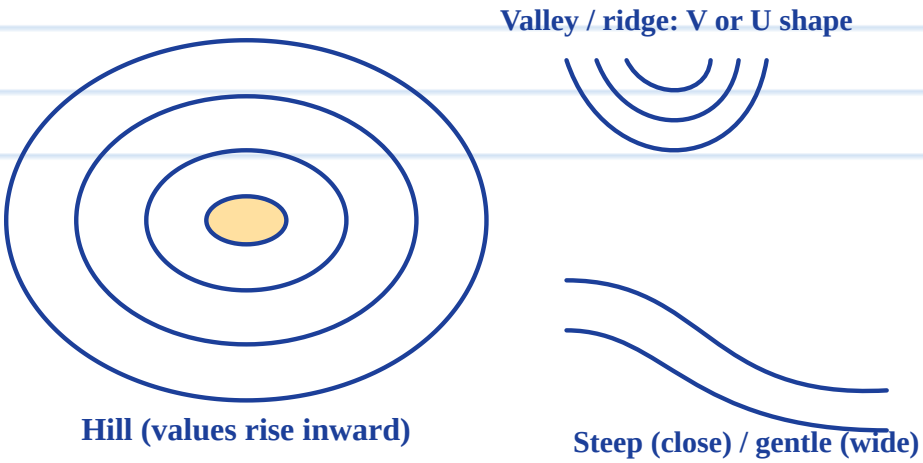


Fig 2.3 Contour patterns – hill, valley, slope

> Contour line: an imaginary line joining points of equal elevation on the ground; contour interval (CI) = vertical distance between successive contours (0.5, 1, 2, 5, 10 m depending on scale & terrain); horizontal equivalent (HE) = horizontal distance between them.

- Contour map = a plan showing relief by contour lines.

❖Characteristics of contours

- Contour lines are closed curves (may close outside the map); do not cross each other (except overhanging cliffs/ caves) and do not merge/ split (except vertical cliff).
- Close spacing → steep slope; wide spacing → gentle slope; equally spaced → uniform slope; straight parallel → plane slope.
- Concentric closed contours with higher values inside → hill; lower values inside → depression/ pond.
- V-shaped / U-shaped contours – pointing to higher ground = valley; pointing toward lower = ridge/ spur.

- Contour is perpendicular to the direction of steepest slope (watershed/ drainage lines cut contours at right angles). Contours cross a ridge or valley line at right angles.
- Ridge line = watershed; valley line = thalweg.
- Gradient = CI / HE (e.g. 1 in n).

❖Methods of contouring

- Direct method: staff is moved till the required level reading is found; points located & plotted – accurate, slow (small areas).
- Indirect method: (a) Grid / squares method – area divided into squares (5–20 m); levels at corners → interpolate contours; (b) Cross-section method – roads/ canals; (c) Radial lines method (tacheometric – hills, from a central station).
- Interpolation of contours: by estimation, arithmetic calculation, or graphical method (tracing paper/ parallel lines).

❖Uses of contour maps

- Selecting site for roads, railways, canals, dams & reservoirs (catchment area, storage capacity), pipelines, airports.
- Finding intervisibility between points, longitudinal sections, earthwork volumes (cut/fill), determination of catchment area & watershed, gradients, planning drainage, military & landscape planning.
- Capacity of reservoir: $V = \Sigma\ (A_1 + A_2)/2 \times CI$ (or prismoidal formula) where A = area enclosed by contour.

❖Example

> Contours 100 m and 102 m are 40 m apart (HE) on plan: gradient = CI/HE = 2/40 = 1 in 20 = 5 %. Contours of 100 m and 105 m at 25 m: slope = 5/25 = 1 in 5 (steep).

9. Additional Short Notes

❖Plane table surveying (brief)

- Surveying + plotting at the same time in the field using a drawing board (plane table) on a tripod, alidade, spirit level, plumb fork, trough compass; methods – radiation, intersection, traversing, resection.

❖Traversing

- A series of connected survey lines; **closed traverse** (polygon) or **open traverse**; angles & lengths measured; **Bowditch rule** for closing error; **Gale's traverse table**.

❖Modern survey tools

- **Total station, GPS/ GNSS (NavIC in India), Drone (UAV) photogrammetry, LiDAR, GIS & remote sensing, Survey of India (Dehradun)** is the national mapping body.

10. More Practice Problems

❖Problem A – Reduce the levels (collimation method)

> Staff readings (m) taken at 20 m intervals with the level fixed at one setup: 0.585 (BS on BM, RL 150.000), 1.015, 1.640, 2.220, 1.915, 0.320 (FS). H.I. = 150.000 + 0.585 = 150.585.

RLs: 149.570, 148.945, 148.365, 148.670, **150.265**. Check: $\Sigma BS - \Sigma FS$ = 0.585 – 0.320 = 0.265 = 150.265 – 150.000 ✔.

❖Problem B – Slope of a line from contours

> Two points lie on contours 220 m and 226 m; plan distance 120 m. Slope = 6/120 = **1 in 20**; gradient = **5 %**; angle = $\tan^{-1}(0.05)$ = **2.86°**.

❖Problem C – Bearings of a closed traverse

> Traverse ABCD, FB of AB = 45°. Interior angles B = 110°, C = 85°, D = 80°. BB of AB (bearing of BA) = 225°. For a clockwise traverse $FB_{next} = BB_{prev}$ – interior angle (add 360° if negative): **FB BC = 225° – 110° = 115°**; BB of BC = 295°; **FB CD = 295° – 85° = 210°**; BB of CD = 30°; **FB DA = 30° – 80° = –50° + 360° = 310°**; BB of DA = 130°; check **FB AB = 130° – 85°** (angle A = 360° – 110° – 85° – 80° = 85°) = 45° ✔.

❖Problem D – Chain correction

> A field measured 600 m with a 30 m chain found 0.1 m too long (30.10 m). True distance = 600 × 30.10/30 = **602.0 m**.

❖Problem E – Curvature & refraction

> Sight length = 300 m = 0.3 km → correction = 0.0673×0.3^2 = 6.06 mm (reading is too high; subtract from staff reading).

11. Errors in Levelling

- **Instrumental:** collimation error (line of sight not parallel to bubble axis) – cancel by keeping **BS = FS distance**; staff graduation errors; bubble sluggish.
- **Personal:** staff not held vertical, wrong reading/ booking, parallax, eye fatigue (staff reading is **least when staff is tilted** → hold vertical & wave slowly to read minimum).
- **Natural:** curvature, refraction, wind, temperature, settlement of tripod.
- **Mistakes:** misreading the staff (e.g. 1.234 as 1.324), booking in wrong column, moving the instrument at a change point.
- Allowable closing error in levelling = $\pm C\sqrt{K}$ mm (K = distance in km): about $\pm 24\sqrt{K}$ for ordinary and $\pm 12\sqrt{K}$ for precise levelling.

12. Comparison Tables

Feature	Chain survey	Compass survey	Theodolite
Measures	Linear only	Bearings	H & V angles
Accuracy	Low	Medium (30′)	High (20″)
Use	Small open flat area	Rough traverse	Precise control, setting out
Local attraction	No	Yes	No
Prismatic vs surveyor's compass	Prismatic	Surveyor's	
Graduation	WCB (0–360°)	QB (0–90°)	
Card	Fixed to the box (needle free)	Fixed to the needle (rotates)	
Reading	Through prism, any distance	Direct through glass lid	

- Quick formulae to remember:
- BB = FB ± 180°; QB ↔ WCB table; True B = Magnetic B ± declination (E +).

- H.I. = RL + BS; RL = H.I. – FS; ΣBS – ΣFS = ΣRise – ΣFall = Last RL – First RL.
- Gradient = CI / HE; Sum of interior angles = (2n – 4) × 90°.

13. Important Questions

- Define surveying; state its objectives and principles.
- Explain chain surveying: instruments, ranging, offsets, errors.
- What is bearing? Distinguish WCB & QB; convert & solve problems on FB, BB, local attraction.
- Define levelling terms (BM, BS, IS, FS, HI, CP, RL). Explain dumpy level & its adjustments.
- Reduce the levels (rise & fall / HI) with check.
- Define contour; explain characteristics, methods & uses of contouring.

Practise at least 4–5 levelling table problems – guaranteed marks!